MILITARY SPECIFICATION

RADIAC SET, AN/PDR-69

1. SCOPE. This specification covers the requirements for the radiac set designated AN/PDR-69. The radiac set includes the IM-213/PD radiacmeter, the DR-351/PD probe the various accessories and a transit case. The radiac set is used for detection of fissionable materials.

2. APPLICABLE DOCUMENTS. The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-B-18/9	Batteries, Dry BA-30
MIL-D-1000	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation of

DRAWINGS

Air Force

69C14343	Audible Indicator
69C14352	Cable, Interconnecting

FSC 6665

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69C14371	Cable, Interconnecting
69D14346	Battery Belt
69D14348	Recorder
69D14324	Case, Radiacmeter and Recorder
69D14337	Radiacmeter, IM-213/PD
69D14334	Probe, DT-351/PD
69B14373	Check Source
69D14384	Transit Case

3. REQUIREMENTS

3.1 PREPRODUCTION. This specification makes provision for preproduction testing.

3.2 DESIGN AND CONSTRUCTION. The radiac set shall be designed and constructed in conformance with Drawings Nos:

69C14343
69C14352
69C14371
69D14346
69D14324
69D14334
69B14373
69D14384

and subsidiary drawings.

3.2.1 WEIGHT. The total weight of the radiac set, including a compliment of 30 dry cells shall not exceed 70 pounds.

3.2.2 BATTERIES. Battery holders shall accept any D-cell specified by MIL-B-18/9.

3.2.3 SCINTILLATION CRYSTALS. Europium-activated calcium fluoride scintillation crystals shall be used. The light output of the crystals shall be no less than 45% of the light output of thallium-activated sodium iodide.

3.2.4 INTERCHANGEABILITY. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not interchangeable, the criteria listed in MIL-D-1000 shall apply and new drawings shall be prepared.

3.2.5 FINISH. Components of the radiac set shall be finished as shown in the drawings.

3.3 PERFORMANCE REQUIREMENTS

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3.3.1 SENSITIVITY. The sensitivity for detection of ^{239}Pu shall be no less than 20 counts per minute (cpm) above noise and background. The sensitivity shall not vary by more than ± 5% for the environmental conditions specified herein.

3.3.2 NOISE AND BACKGROUND. Noise and background shall not exceed 2000 cpm when the instrument is adjusted to have a threshold at 6-7 KeV and a window-width of 22 KeV and operated in an environment free of x and/or gamma radiations from artificial sources and/or radioactive contamination.

3.3.3 RESOLUTION. The resolution (FWHM) of the instrument for the 17 KeV x-ray peak due to a ^{241}Am source shall not exceed 100%.

3.3.4 GAIN CONTROL. When the gain control is adjusted to cause the pulse-height analyzer controls to read directly in energy, the gain control setting shall not be less than 5.0 or greater than 80. (The gain control dial reads 0 to 100).

3.3.5 ANALYZER CONTROLS. When the gain control is adjusted to cause the controls for analyzer A to read directly in energy, the controls of analyzer B shall read within ± 5% of true energy.

3.3.6 COUNTING INTERVAL. The high range counting interval shall be 4.80 ± 0.05 sec.

3.3.7 RANGING. There shall be a factor of 100 ± 2 between ranges.

3.3.8 STRIP CHART RECORDER. Each channel of the recorder shall have a four-decade logarithmic response extending from ∿ 60 c/min to 6000,000 c/min.

3.4 ENVIRONMENTAL REQUIREMENTS. The radiac set (recorder excluded) shall meet performance requirements during or after, as applicable, exposure to the following environmental conditions:

a. Low temperature to -40°F
b. High temperature to + 120°F
c. Altitude to 20,000 ft
d. Temperature shock, 32°F - 120°F
e. Solar radiation
f. Rain when exposed under service conditions
g. Humidity, incident to storage, transportation and handling in a damp climate
h. Fungus such as encountered in storage or use in tropical climates
i. Vibration as incident to transportation and field use
j. Shock as incident to transportation and field use

3.5 OPERATIONAL LIFE. When loaded with high-quality zinc-carbon D-cells (Eveready 1150 or equivalent) and operated at any temperature in the range from 70°F to 140°F, the radiacmeter and probe shall meet performance requirements for a total operating time of 52 hours with the duty cycle specified below. With a temperature of 32°F, and other conditions as stated above, the

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total operating time shall be 30 hours.

When operated with the auxiliary battery pack, at any temperature in the range from -40°F to +120°F, with other conditions as stated above, the operating time shall be 52 hours; the temperature of the auxiliary battery pack shall be above 65°F.

The duty cycle is to be 8 hours on, 8 hours off, with a final on period to give the total on time stated above.

3.6 IDENTIFICATION OF PRODUCT. The radiac set shall be marked for identification in accordance with MIL-STD-130.

3.7 WORKMANSHIP. Particular attention shall be given to installation of seals and gaskets, and the quality of plate-through holes in printed circuit boards.

4. QUALITY ASSURANCE PROVISIONS.

4.1 CLASSIFICATION OF TESTS. Inspection and testing shall be classified as follows:

a. Preproduction Tests
b. Acceptance Tests

4.2 PREPRODUCTION TESTING.

4.2.1 PREPRODUCTION TEST SAMPLES. The preproduction test samples shall be representative of the production radiac sets. They shall be tested at a laboratory designated by the procuring activity, or when so stated in the contract, at the contractor's plant, under supervision of the procuring activity.

4.2.1.1 PREPRODUCTION TEST REPORTS. In the event that preproduction testing is performed at the contractor's plant, the contractor shall, upon completion of the tests, prepare a test report in accordance with MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.2.2 PREPRODUCTION TESTS. Preproduction tests shall include all tests scheduled in Table I.

4.3 ACCEPTANCE TESTS. Acceptance tests shall include the following tests as scheduled in Table I.

a. Individual Tests
b. Sampling Tests

4.3.1 SAMPLING PLAN. The sampling plan and sampling lot sizes shall be in accordance with MIL-STD-105. Level II inspection of Table V(A) shall be

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used. The AQL shall be (2.5). A sampling test plan shall be submitted to the procuring activity for approval before production begins.

4.3.2 REJECTION AND RETEST. When a sample item fails to meet the specified tests, no item still on hand or later produced shall be accepted until the extent and cause of failure are determined and corrected.

4.3.3 INDIVIDUAL TESTS MAY CONTINUE. Individual tests may be continued pending an investigation of a sampling test failure. However, final acceptance of items on hand or produced later shall not be made until it is determined that the items meet all requirements of this specification.

4.3.4 DEFECTS IN ITEMS ALREADY ACCEPTED. The investigation of a test failure could indicate that defects may exist in items already accepted. In such a case, the contractor shall advise the procuring activity of such possible defects and the means of correcting them.

4.4 TEST CONDITIONS. Test conditions shall be as specified for the tests scheduled in Table I.

4.5 TEST METHODS.

4.5.1 EXAMINATION OF PRODUCT. Each component of each radiac set shall be examined for compliance with the requirements of this specification for which no specific tests are scheduled. Among such requirements are fit and finish, workmanship and completeness of the radiac set. Failure to satisfy these requirements shall be cause for rejection.

4.5.2 CONFORMANCE TO DRAWINGS. Each component of each radiac set shall be inspected for strict conformance to drawings and all standards and specifications referenced in the drawings. Failure to satisfy these criteria shall be cause for rejection.

4.5.3 WEIGHT. The radiac set (including 30 D-cells, but no spare parts) shall be weighed by any convenient method. Weight over 70 pounds shall be cause for rejection.

4.5.4 BATTERY COMPATIBILITY. The depth and diameter of the battery wells in the radiacmeter and the recorder shall be measured with vernier calipers to verify compatibility with all D-cells (BA-30) specified by MIL-B-18/9. Dimensional incompatibility is cause for rejection.

4.5.5 SCINTILLATION CRYSTALS. Demonstration of satisfactory resolution of the L X-ray peak in the pulse height spectrum due to a ^{241}Am source shall be considered evidence of adequate light output. The crystal shall be mounted in a probe for this test.

Place a 3.00 μCi ^{241}Am source (Dwg 69B14373) on a horizontal nonmetallic work surface. Stand the probe upright over the source with the source

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TABLE I

Test	Requirement Paragraph	Paragraph	Prod.	Individual	Sampling
Weight	3.2.1	4.5.3	x		
Examination of Product	3.2.2,3.2.5, 3.6,3.7,4.6	4.5.1	x	x	
Conformance to Drawings	3.2.2,3.2.4	4.5.2			
Scintillation Crystals	3.2.5	4.5.4	x		
Sensitivity	3.2.2	4.5.5	x		x
Noise and Background	3.3.1	4.5.6	x	x	
Resolution	3.3.2	4.5.7	x	x	
Gain Control	3.3.3	4.5.8	x	x	
Analyzer Controls	3.3.4	4.5.9	x	x	
Counting Interval	3.3.5	4.5.10	x	x	
Ranging	3.3.6	4.5.11	x	x	
Strip Chart Recorder	3.3.7	4.5.12	x	x	
Environmental	3.3.8 3.4	4.5.13 4.6	x	x	
Low Temperature	a	4.6.1	x		
High Temperature	b	4.6.2	x		
Altitude	c	4.6.3	x		
Temperature Shock	d	4.6.4	x		
Solar Radiation	e	4.6.5	x		
Rain	f	4.6.6	x		
Humidity	g	4.6.7	x		
Fungus	h	4.6.8	x		
Vibration	i	4.6.9	x		
Shock	j	4.6.10	x		
Operational Life	3.5	4.6.11	x		x

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approximately centered under the beryllium window (positioning is not critical).

With laboratory equipment of the type suitable for low-energy X-or gamma-ray scintillation spectroscopy, including a multichannel analyzer, record the differential pulse-height spectrum of signals from the probe. Determine the resolution of the L X-ray peak in the usual way. Resolution in excess of 100% (FWHM) is cause for rejection.

4.5.6 SENSITIVITY. Sensitivity shall be measured by the methods described here.

4.5.6.1 SENSITIVITY MEASUREMENT, METHOD I. Place a 3.00 μCi ^{241}Am source (Dwg 69B14373) on a horizontal work surface. Locate the probe 30 cm above the source, with the beryllium window centered on a vertical through the source. Select Channel A. Alternately run differential pulse-height spectra and adjust gain to make the Channel A analyzer controls read directly in energy when the analyzer is considered to cover 0 to 100 KeV in 10 turns. For final adjustment, set the Channel A window to 2.0 KeV, and carefully measure the spectrum, with further fine adjustments of the gain as necessary. Set the Channel A threshold at the mid-point of the minimum between the electronic noise and background, and the lower side of the L X-ray peak. Set the Channel A window to 22 KeV. Record gain and Channel A control settings for future reference.

Observe the gross count (noise and background plus signal), taking enough counts to assure 1% counting statistics. Remove the source, and observe the count due to noise and background, taking enough counts to assure 1% counting statistics. The signal count rate, (noise and background plus signal) - (noise and background), shall be no less than 20 counts per minute. A signal count rate less than 20 counts per minute shall be cause for rejection.

4.5.6.2 SENSITIVITY MEASUREMENTS, METHOD II. For certain tests, it may not be practicable to locate the source as described in Method I. Method II may be applied in such cases.

Complete the procedure for Method I and record the sensitivity. Leave gain and analyzer control settings unchanged. Place source and probe in the positions they will occupy for test purposes, and determine the count rate due to the source. If R_t is the rate observed in a test measurement, the corresponding sensitivity is

$$S_t = Rt \times \frac{S_o}{R_o}$$

Where S_o is the sensitivity determine by Method I, and R_o is the pre-test count rate.

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If, during or after a test, the sensitivity cannot be restored to $S_o \pm 5\%$, by adjusting the gain control, the test shall be considered a failure.

4.5.7 NOISE AND BACKGROUND. Combined electronic noise and background shall be measured with the conditions stated in Para. 4.5.6.1, but with the check source removed. Combined noise and background is excess of 2000 counts per minute is cause for rejection.

4.5.8 RESOLUTION. Resolution (for the probe and radiacmeter in combination) shall be determined graphically from a differential pulse-height spectrum obtained by the method of Para. 4.5.6.1, with the source in any convenient position. Resolution in excess of 100% FWHM for the L X-ray peak shall be cause for rejection.

4.5.9 GAIN CONTROL. Observe gain control setting upon completion of procedures outlined in Paras. 4.5.6.1 and/or 4.5.6.2. A gain control setting of less than 5.0 or more than 80.0 is cause for rejection.

4.5.10 ANALYZER CONTROLS. Set the gain control to make the channel A analyzer controls read directly in energy, by the procedure given in Para. 4.5.6.1. Select Channel B, and run a differential pulse-height spectrum with a 2.0 KeV window. The pulse-heights at which the L X-ray and 60-KeV peaks appear in the Channel B spectrum shall agree within $\pm$ 5% with the pulse heights at which the respective peaks appear in the Channel A spectrum; failure to achieve $\pm$ 5% agreement is cause for rejection.

4.5.11 COUNTING INTERVAL. Feed +10V 100 µsec wide pulses with a repetition rate of N per sec into pin "B" of the recorder connector on the radiacmeter. The repetition rate, N, may be between 1000/sec and 2000/sec and must be known to $\pm$ 0.1% or better. Select the high range. The count should be 30 N. A count that differs by more than $\pm$ 1% from 30 N is cause for rejection.

4.5.12 RANGING. Select the low range. Feed in pulses as described in Para. 4.5.11, but with a repetition rate of N/100. The low range indication shall be 30 N $\pm$ 2%. Any other indication shall be cause for rejection.

4.5.13 STRIP CHART RECORDER. Feed +10V 100 µsec wide pulses into pin B of the input connector on the recorder unit.

Recorder readings shall be

% full scale	pulses/minute
0	60 $\pm$ 10%
25	600 $\pm$ 10%
50	6000 $\pm$ 10%
75	60,000 $\pm$ 10%
100	600,000 $\pm$ 10%

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Repeat the measurements for the other channel by feeding pulses to pin D of the input connector.

Input pulse rates different from those given in the table above shall be cause for rejection.

4.6 ENVIRONMENTAL TESTS. During or after subjection to the environmental tests, as specified in the test, the test set shall be subjected to the functional tests. Failure to satisfy the requirements of the functional tests shall be considered as failure to pass the environmental tests and cause for rejection. Sensitivity shall be measured by either Method I or Method II, as appropriate, before, during, and after tests. No other measurements are specifically required.

4.6.1 LOW TEMPERATURE. The low temperature test shall be performed in accordance with MIL-STD-810, Method 502, Procedure I, with the following exceptions:

a. Step 2. The equipment shall be in the closed transit case. The temperature shall be -40°F for 48 hours.

b. Step 3. The temperature shall be -40°F for 24 hours, the air velocity shall be 20mph. The auxiliary battery pack shall not be cooled.

c. Step 4. Use the auxiliary battery pack, Method II, as convenient. The radiacmeter, probe, audible indicator auxiliary battery pack shall be tested in combination.

4.6.2 HIGH TEMPERATURE. The high temperature test shall be performed in accordance with MIL-STD-810, Method 501, Procedure I, with the following exceptions:

a. Step 3. The equipment shall be in the closed transit.

b. Step 5. The instrument shall be operated for 24 hours.

The high temperature shall be 130°F in all phases of the test.

4.6.3 ALTITUDE. The altitude test shall be performed in accordance with MIL-STD-810, Method 500, Procedure I, with the following exceptions:

a. Step 2. The equipment shall be in the closed transit case. Pressure shall be maintained for 3 hours.

b. Pressure shall be that for 20,000 ft. altitude.

4.6.4 TEMPERATURE SHOCK. The temperature shock test shall be performed in accordance with MIL-STD-810, Method 503, Procedure I, with the following exceptions:

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a. Step 1. The equipment shall be in the closed transit case. The temperature shall be 120°F.

b. Step 2. The temperature shall be 32°F.

c. Step 4. The temperature shall be 120°F.

4.6.5 SOLAR RADIATION. The solar radiation test shall be performed in accordance with MIL-STD-810, Method 505, Procedure I. Sensitivity shall be monitored during the test.

4.6.6 RAIN. The rain test shall be performed in accordance with MIL-STD-810, Method 506, Procedure I, with the following procedures:

a. The test shall be conducted with the instrument in the closed transit case.

b. The test shall be conducted with the probe, radiacmeter, recorder, and audible indicator removed from the transit case.

4.6.7 HUMIDITY. The humidity test shall be performed in accordance with MIL-STD-810, Method 507, Procedure IV.

4.6.8 FUNGUS. The fungus test will be performed in accordance with MIL-STD-810, Method 508, Procedure I. The probe will not be opened for inoculation.

4.6.9 VIBRATION. The vibration test will be performed in accordance with MIL-STD-810, Method 514, Procedure XI.

4.6.10 SHOCK. The shock test will be performed in accordance with MIL-STD-810. Method 516, Procedure II and V. The probe will not be subjected to the Procedure V test.

4.6.11 OPERATIONAL LIFE. The radiacmeter, probe and audible indicator shall be tested in combination. The auxiliary battery pack shall not be tested. Set-up, pre-test and post-test conditions are generally as stated in Para. 4.6.2.

Equipment shall be off. Soak at 120°F for 4 hours. At 4 $\pm$ 1/2 hr, turn instrument power on. Determine sensitivity by Method II, if count is out of tolerance, the gain control only, may be adjusted to restore calibration. Leave instrument power on.

At 12 $\pm$ 1/4 hr, repeat the sensitivity measurement. Turn power off.

At 20 $\pm$ 1/4 hr, turn power on, repeat the sensitivity measurement. Leave power on.

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Continue alternate power-on, power-off cycles, with sensitivity measurements, to 100 hr. At 100 hr, measure sensitivity. Turn off power. Allow equipment to cool to room temperature. Measure sensitivity.

Causes for rejection are:

i. out-of-tolerance sensitivity which cannot be corrected by adjusting the gain control only.

ii. failure to satisfy other performance specifications stated herein.

iii. any other malfunction, or deterioration which would compromise performance in future.

5. PREPARATION FOR DELIVERY.

5.1 PRESERVATION AND PACKAGING.

a. All batteries shall be removed from equipment for shipping, and during storage prior to shipment.

b. The "peck bar" of the recorder shall be down for shipment.

c. No special packaging or preservation is required.

5.2 PACKING. Equipment shall be packed in the transit case (Dwg 69D14384).

5.3 MARKING. The transit case shall be marked in accordance with MIL-STD-129. The nomenclature shall be

RADIAC SET, AN/PDR-69

6. NOTES

6.1 ORDERING DATA. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Conditions under which government-owned property will be made available to the contract.

c. Authorization for recipients of shipments to possess radioactive material of the type and quantity in the check sources.

d. Whether preproduction tests will be conducted by the procuring activity.

6.1.1 PREPRODUCTION TEST SAMPLES. It is expected that the contract or purchase order will specify that a minimum of units representative of the

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production charger will be presented for test and evaluation. The invitation for bids, and the contract or purchase order should specify the point of inspection for these tests.

6.2 PRODUCTION DELIVERY. No production item should be delivered prior to approval of the preproduction samples.

6.3 DEFINITIONS

a. PREPRODUCTION TESTS. Preproduction tests are those tests performed prior to production on representative samples of the production equipment to demonstrate that the equipment meets the requirements of this specification.

b. ACCEPTANCE TESTS. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

CUSTODIAN
Air Force - 15

PREPARING ACTIVITY
Air Force - 15

Project No. 6665-F295

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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